



NETWORK SECURITY | MODULE 1

Foundations of Network Security

Lesson 1 – Introduction to Network Security
Lesson 2 – Network Models



Cryptography · Attacks · Services · Mechanisms · LAN/MAN/WAN · OSI & TCP/IP



LESSON 1

1

Why Security?

The information age and the CIA goals

2

Cryptology Basics

Cryptography vs. cryptanalysis

3

Attacks, Services, Mechanisms

Three ways to look at security

4

Security Attacks

Four ways to attack information

LESSON 1 (cont.) → LESSON 2

5

Cryptographic Systems

Substitution, transposition, keys, blocks

6

Attacking Cryptosystems

Brute force and cryptanalysis

7

Networks & Categories

LAN, MAN, WAN, internetworks

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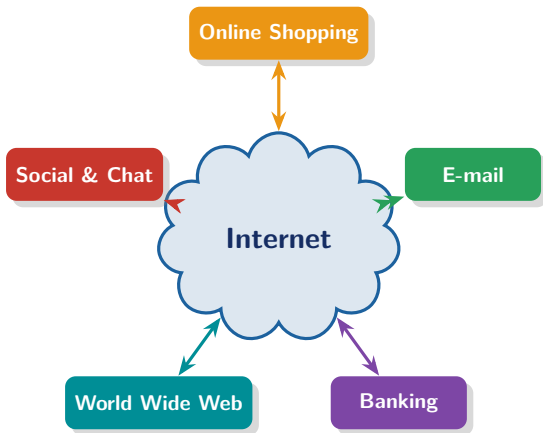
Layered Models

Postal analogy, TCP/IP, OSI, IAB



Lesson 1

Introduction to Network Security

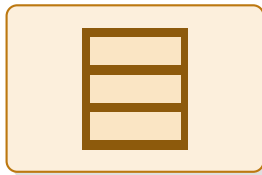


- The Internet has changed daily life: **shopping, e-mail, WWW, banking**
- This convenience created **dependency** on the Internet
- As an *open forum*, the Internet also created security problems

People need to be sure that...

- their communication stays **confidential**
- online vendors are **authentic**
- bank requests keep their **integrity**

Era 1: Physical Files



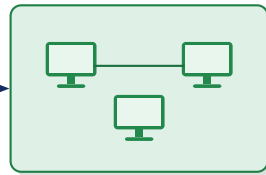
Lock the cabinet;
trusted people only

Era 2: Computers



Stored electronically;
same requirements

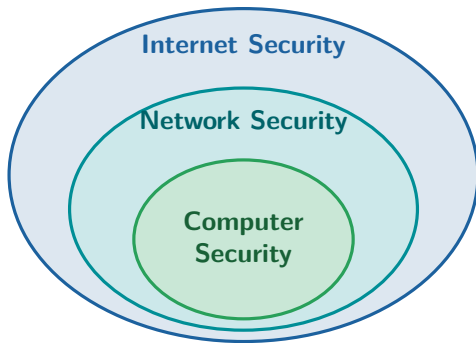
Era 3: Networks



Information is **distributed**
and **transmitted**

Key idea

Information is an **asset** with value like any other asset. It must be **hidden** from unauthorized access, **protected** from unauthorized change, and **available** to authorized users. The requirements stayed the same, but in networks they must hold *both at rest and in transit*.



Computer security Tools that protect data stored on a computer and stop hackers

Network security Measures that protect data **during transmission**

Internet security Measures to **prevent, detect and correct** violations across a collection of interconnected networks

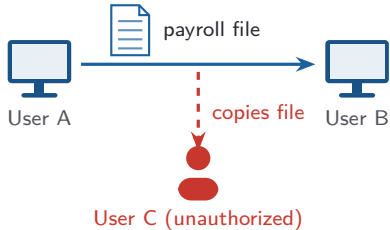
No clear boundaries

A virus may arrive on a disk *or* over the network. Once it is resident, computer-security tools must detect and remove it.

Two Examples of Security Violations

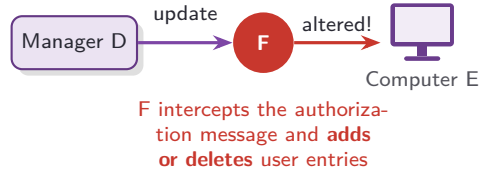


Example 1: Disclosure



C **monitors** the transmission and **captures** a copy of sensitive data.

Example 2: Alteration

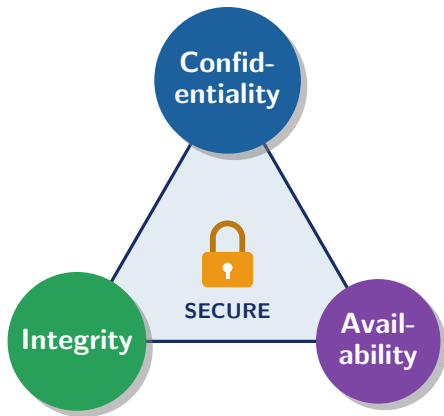


A forged **authorization file** gives access to the wrong people.

Takeaway

Network security means protocols that let us use the Internet **comfortably**, without worrying about such attacks.

The Three Security Goals: The CIA Triad



- **Confidentiality:** hide information from *unauthorized access*
- **Integrity:** protect information from *unauthorized change*
- **Availability:** information is available to *authorized entities when needed*

Remember

Every attack threatens at least one of these three goals.



Confidentiality

Question

Who can **read** it?

Example

A bank keeps your balance secret

Threatened by

Eavesdropping, traffic analysis

Integrity

Question

Has it been **changed**?

Example

A cheque amount cannot be altered

Threatened by

Modification, fabrication

Availability

Question

Can I **use** it now?

Example

An ATM works at 2 a.m.

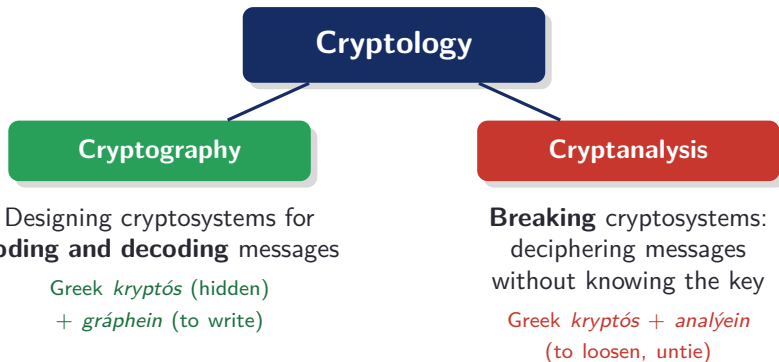
Threatened by

Denial of service

Cryptology = Cryptography + Cryptanalysis



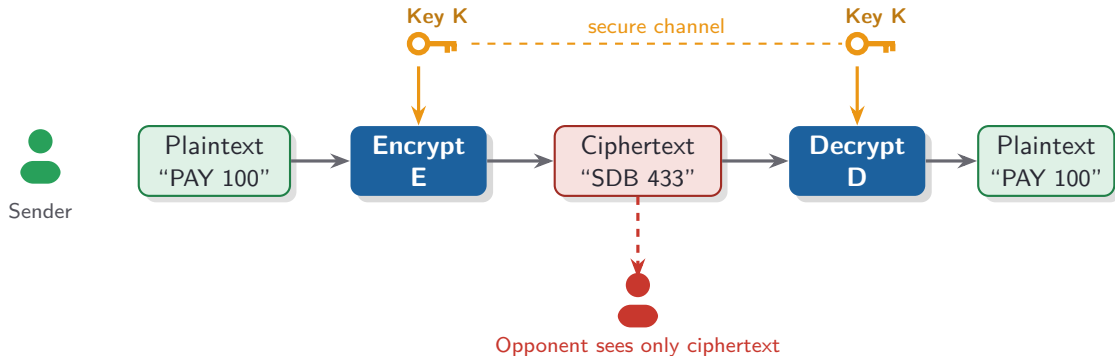
Greek *kryptós* “hidden” + *logos* “word”: the science of secret messages



Common mistake

Cryptology is often wrongly treated as a synonym for cryptography. It is the **broader** term that covers both.

The Basic Encryption Model



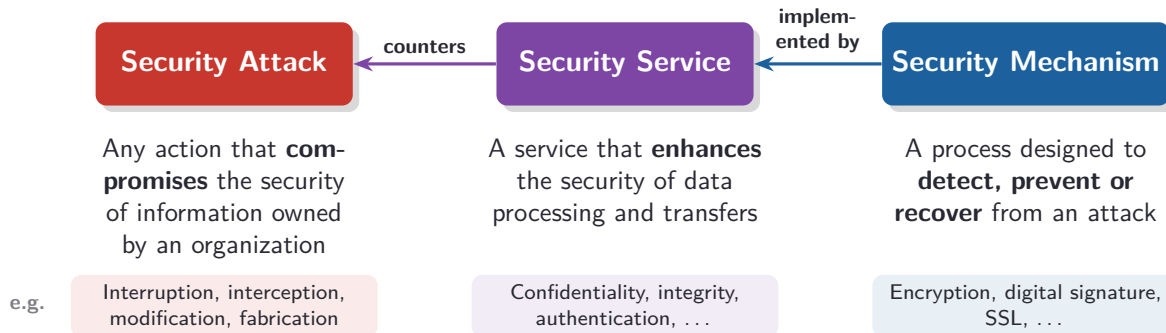
Plaintext: the original readable message

Ciphertext: the scrambled, unreadable message

Cipher: the algorithm (E and D)

Key: the secret value that controls the cipher

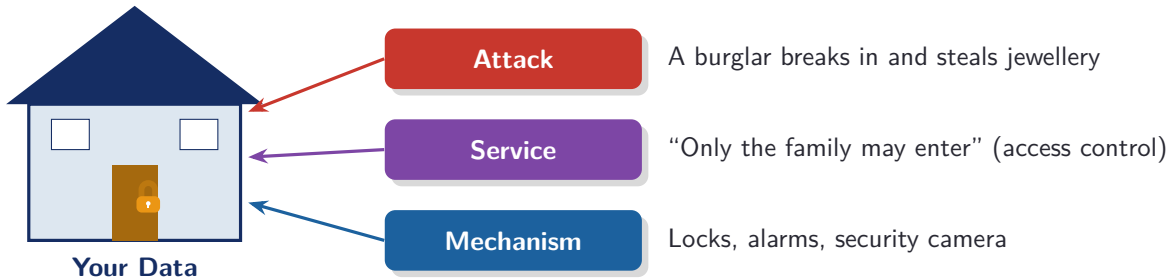
Three Aspects of Information Security



Why these three?

They help an organization **identify its security needs** and **evaluate** security products and policies.

An Analogy: Securing a House



Attack

A burglar breaks in and steals jewellery

Service

“Only the family may enter” (access control)

Mechanism

Locks, alarms, security camera

Mapping to networks

Burglar = hacker · “Only family may enter” = access control / confidentiality service · Lock = encryption mechanism



Security service	What it guarantees	Typical mechanism
Confidentiality	Only authorized parties can read data	Encryption
Integrity	Data is not altered in transit	Hash / message digest, digital signature
Authentication	The sender is who they claim to be	Digital signature, passwords
Non-repudiation	The sender cannot deny sending	Digital signature
Access control	Only permitted users use a resource	Access-control lists, passwords
Availability	The service stays usable	Redundancy, filtering

Key point

A service is **what** we want; a mechanism is **how** we get it. One service may use **one or more** mechanisms.



- Information flows from an **information source** (a file or a user)...
- ...to an **information destination** (another file or user)
- Nobody interferes: this is the **normal** case

Four general categories of attack

Each attack below **changes** this picture in a different way.

Interruption

Interception

Modification

Fabrication



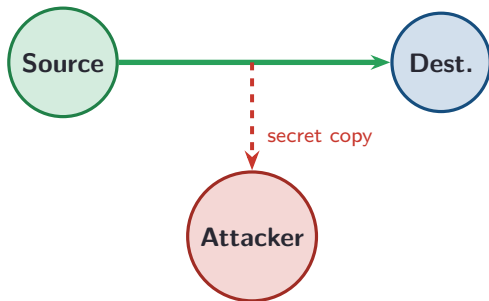
Definition

An asset of the system is **destroyed** or becomes **unavailable** or unusable.

Threatens: **Availability**

Examples

- Cutting a communication line
- Destroying a hard disk
- Disabling a file-management system
- Flooding a server (**denial of service**)



Definition

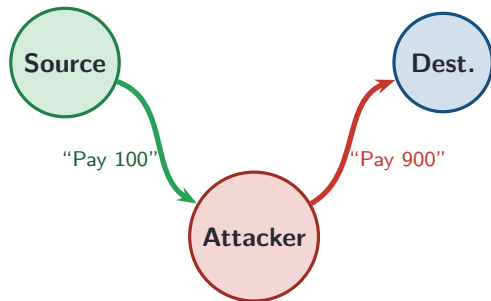
An **unauthorized party** gains access to an asset.

Threatens: **Confidentiality**

Examples

- Wiretapping to capture data
- Illicit copying of files or programs
- Sniffing passwords on a Wi-Fi network

The original message still arrives, so this attack is hard to notice.



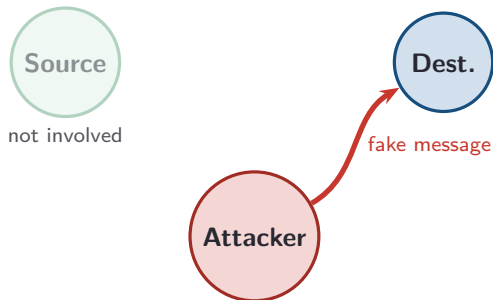
Definition

An unauthorized party not only gains access but also **tamper**s with the asset.

Threatens: **Integrity**

Examples

- Changing values in a data file
- Altering a program so it behaves differently
- Modifying the content of a message in transit



Definition

An unauthorized party **inserts counterfeit objects** into the system.

Threatens: **Authenticity** (integrity of origin)

Examples

- Inserting spurious messages into a network
- Adding a fake record to a file
- Phishing e-mail “from your bank”



Passive Attacks

Goal: obtain information

Nature: eavesdropping, monitoring

Data changed? No

Examples: release of message contents, traffic analysis

Hard to detect, so focus on **prevention** (encryption)

vs

Active Attacks

Goal: change data or harm the system

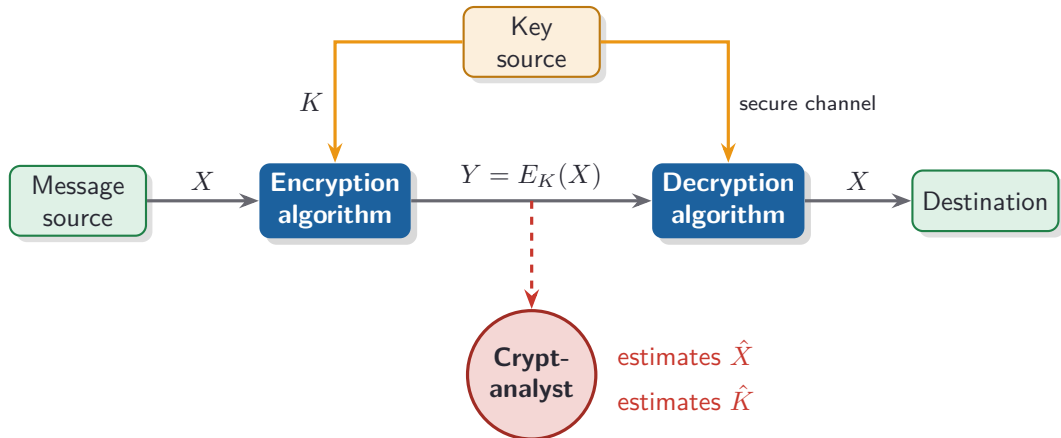
Nature: modification of the data stream

Data changed? Yes

Examples: masquerade, replay, modification, denial of service

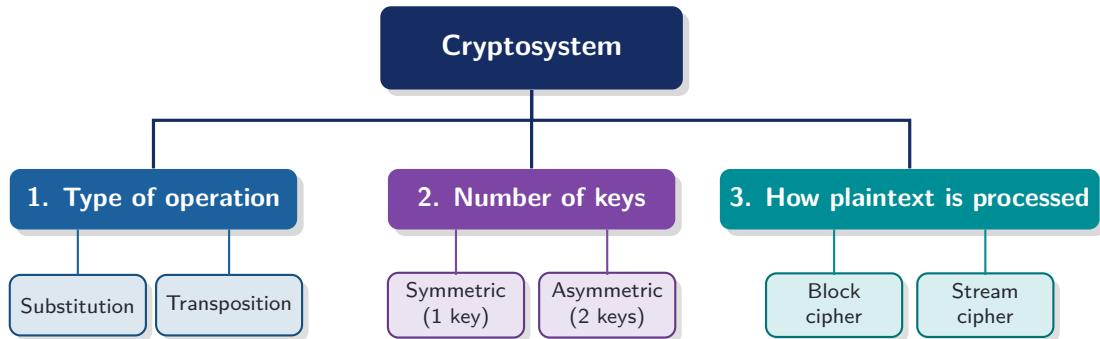
Hard to prevent, so focus on **detection and recovery**

Model of a Conventional Cryptosystem



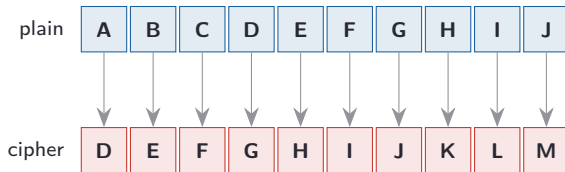
- The opponent sees Y but not K or X , and is assumed to **know the algorithms** E and D
- Recovering $\hat{X}$ breaks **one message**; recovering $\hat{K}$ breaks **all past and future messages**

Three Dimensions of Cryptographic Systems

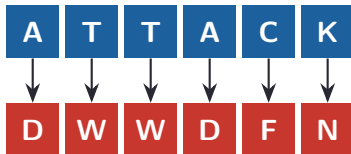


Golden rule

Every operation must be **reversible**: no information may be lost. Most real systems (*product systems*) combine **many stages** of substitution and transposition.



... and so on, wrapping X, Y, Z to A, B, C



Substitution

Each element (bit, letter, group of letters) is **mapped to another element**.

Example: Caesar cipher (shift 3)

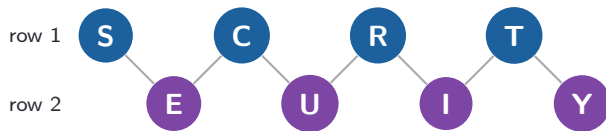
- Each letter moves 3 places forward
- ATTACK → DWWDFN
- Letters **change**, positions **stay**

Weakness

Only 25 possible shifts, so it is easily broken by trying them all.



Rail-fence cipher, depth 2



S C R T E U I Y

ciphertext = row 1 + row 2

Transposition

Elements of the plaintext are **rearranged**: same letters, **new order**.

Example

- Write SECURITY in zig-zag on 2 rows
- Read row by row
- SECURITY → SCRTEUIY

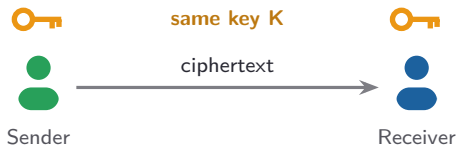
Combine both!

Modern ciphers alternate substitution *and* transposition in many rounds.

Dimension 2: Number of Keys

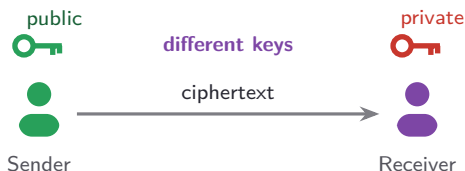


Symmetric (single-key, secret-key)



Also called **conventional** encryption. Fast, but the key must be **shared secretly** first.

Asymmetric (two-key, public-key)



Encrypt with the receiver's **public** key; only their **private** key decrypts. No secret key to share.

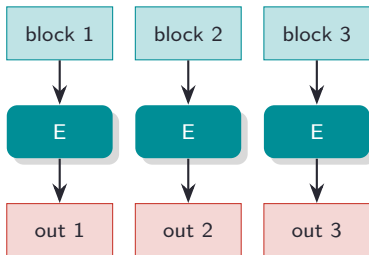
Coming later in the course

Symmetric: DES, IDEA, AES (Modules 6–7) · Asymmetric: RSA, Diffie–Hellman, ECC (Module 8)

Dimension 3: Block vs. Stream Ciphers

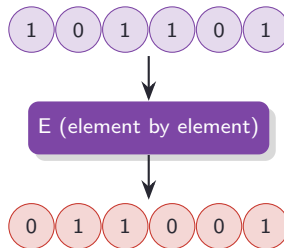


Block cipher



Processes the input **one block at a time** (e.g. 64 or 128 bits), producing one output block per input block.

Stream cipher



Processes input elements **continuously**, producing output **one element at a time** as it goes along.

Analogy

Block = sealing letters in envelopes one by one · Stream = water through a filter, drop by drop

Two Ways to Attack a Cryptosystem



Goal: recover the key

Cryptanalysis

Uses the **nature of the algorithm**, knowledge of plaintext characteristics, or known plaintext-ciphertext pairs to **deduce** the key or plaintext.

"Think smart."

Brute-force attack

Tries **every possible key** until an intelligible plaintext appears. On average **half** of all keys must be tried.

"Try everything."

If either succeeds...

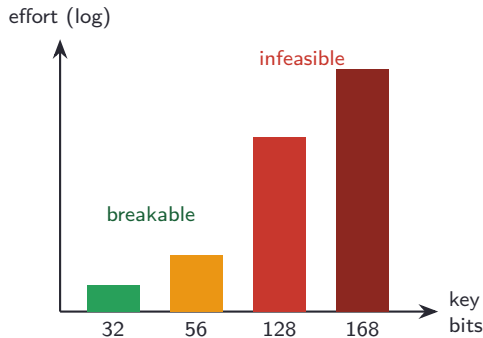
...all future and past messages encrypted with that key are compromised.



Brute Force: Why Key Length Matters

Key	Keys	@ $10^6/s$	@ $10^{12}/s$
32 bits	4.3×10^9	35.8 min	2.15 ms
56 bits	7.2×10^{16}	1142 years	10 hours
128 bits	3.4×10^{38}	5.4×10^{24} yrs	5.4×10^{18} yrs
168 bits	3.7×10^{50}	5.9×10^{36} yrs	5.9×10^{30} yrs

Average time = trying half the key space.



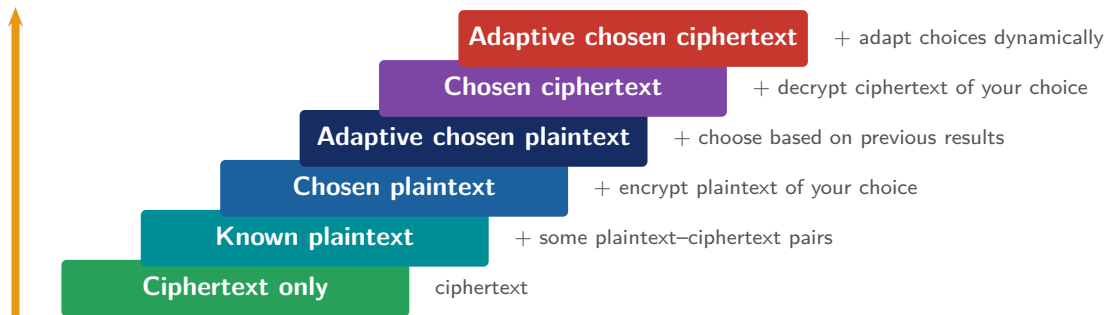
Lesson

Each extra bit **doubles** the work. A 128-bit key makes brute force hopeless, so attackers turn to **cryptanalysis**.

Types of Cryptanalytic Attacks



more attacker
power



The cryptanalyst's strategy depends on **how much information is available**. Going up the stairs, the attacker knows more and the attack gets easier.



Ciphertext only

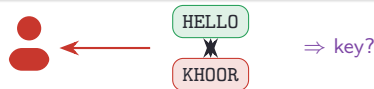
The cryptanalyst has only a **sample of ciphertext**, without the plaintext.



- Data is **easy to obtain**
- Attack is **hard**: needs a very large sample

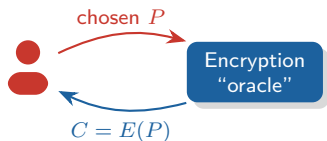
Known plaintext

The cryptanalyst has a sample of ciphertext **and the corresponding plaintext**.



- E.g. standard headers like “Dear Sir”
- Pairs reveal **patterns** in the key

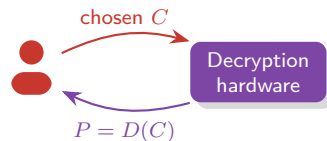
Chosen-Plaintext and Chosen-Ciphertext Attacks



Chosen plaintext

The attacker **chooses** plaintext and obtains its ciphertext.

Adaptive: choices change *dynamically* based on earlier results.



Chosen ciphertext

The attacker chooses ciphertext and obtains the decrypted plaintext. Mostly used against **public-key** systems.

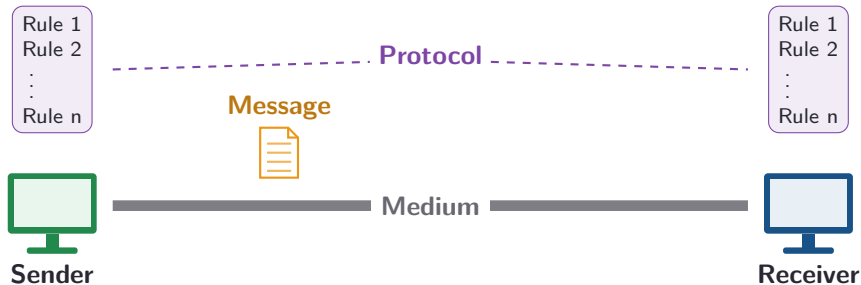
Adaptive: free use of decryption hardware, but cannot extract the key directly.



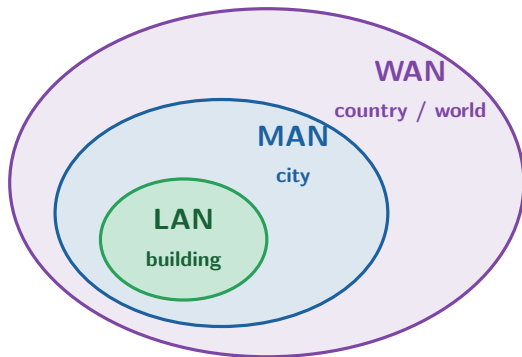
Lesson 2

Network Models

Why Networks? Data Communication



- Data communication means conveying a message, an idea, a picture or speech that is **received and understood correctly**
- The telephone is instant, but **pictures and large data** cannot be sent reliably or remembered
- So sending data over **wired and wireless networks** (messages, pictures, voice) has become vital

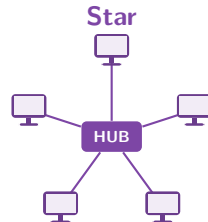
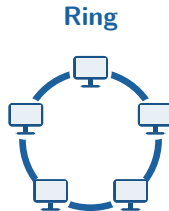
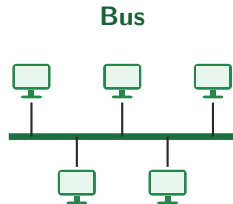


Three primary categories:

- **Local Area Network (LAN)**
- **Metropolitan Area Network (MAN)**
- **Wide Area Network (WAN)**

Category is decided by

size · ownership · distance covered ·
physical architecture

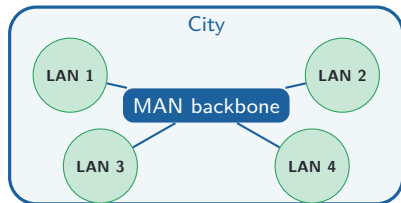


- Usually **privately owned**
- Links devices in one **office, building or campus** (a few km)
- Shares **hardware** (printers), **software** and **data**

- Usually one type of transmission medium
- Common topologies: **bus, ring, star**
- Speeds: 4–16 Mbps traditionally, now **100 Mbps to gigabit**

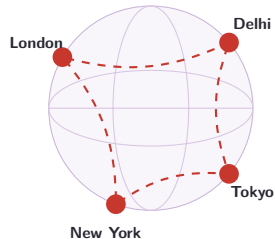


MAN: spans a city



- Connects the LANs of a company's offices across a city
- Private, or leased from a telephone / cable company
- E.g. **SMDS**, cable TV networks

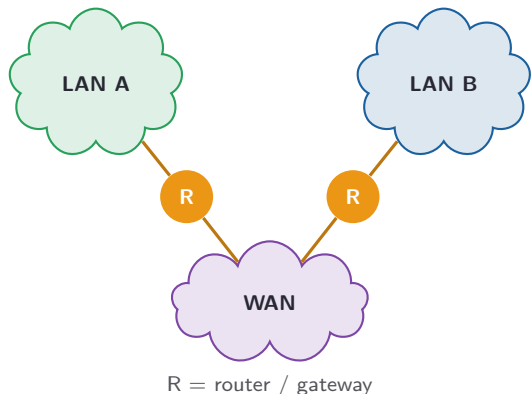
WAN: country, continent, world



- Long-distance transmission of data, voice, image, video
- Uses public, leased or private **communication devices**
- Wholly owned by one company: an **enterprise network**



Feature	LAN	MAN	WAN
Coverage	Office, building, campus	Entire city	Country, continent, world
Distance	A few km	Tens of km	Unlimited
Ownership	Private	Private or public company	Public, leased or private
Speed	High (100 Mbps–Gbps)	Medium–high	Lower, varies
Hardware	Own cabling	Cable / telephone lines	Communication carriers
Example	College computer lab	City-wide cable network	The Internet backbone



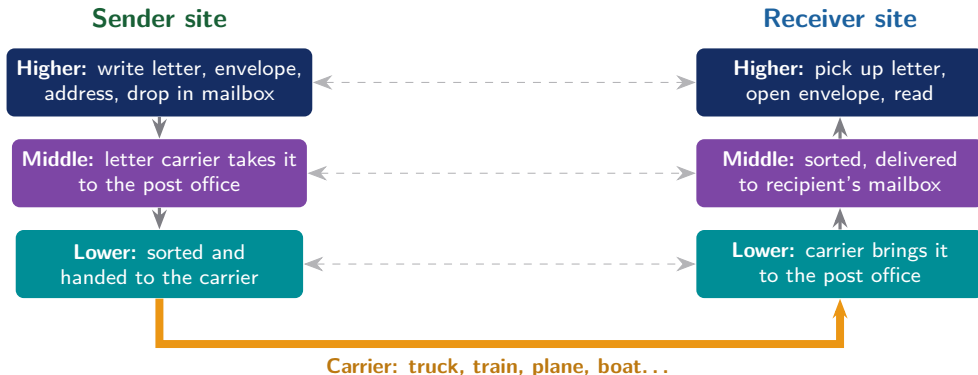
- When two or more networks are connected, they form an **internetwork**
- They are joined by **routers and gateways**

Do not confuse

internet (lowercase *i*): any interconnection of networks

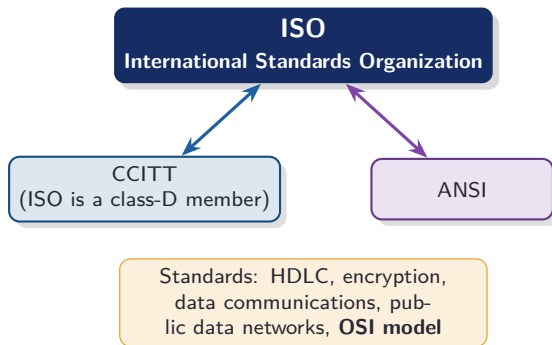
Internet (uppercase *I*): the specific **worldwide** network

Layered Tasks: The Postal Analogy



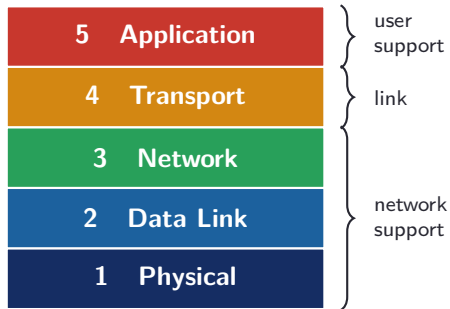
Each layer at the sender **uses the services of the layer below it**. The receiver does the tasks in **reverse order**.

The ISO: Who Makes the Standards?



- A **voluntary organization** of national standards committees from each member country
- Coordinates its activities with **CCITT** on common issues
- Works with CCITT and **ANSI** through subcommittees and working groups
- Best-known product: the **Open Systems Interconnection (OSI)** model

The Internet Model (TCP/IP Protocol Suite)

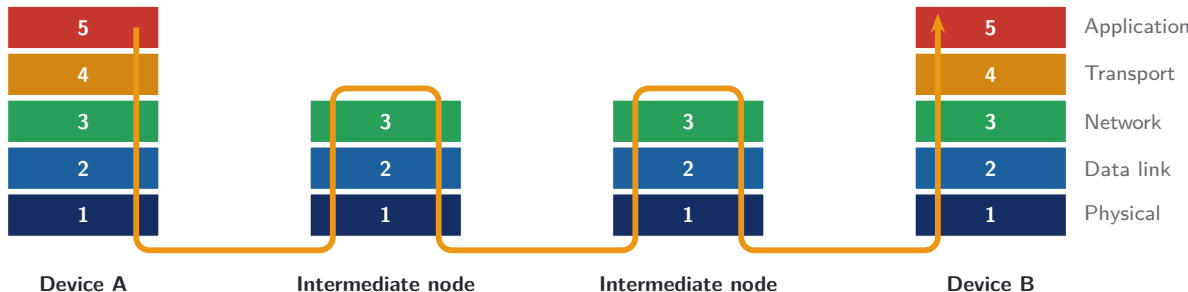


- A **five-layer** model; also called the **TCP/IP protocol suite**
- Each layer defines a **family of functions** distinct from the others
- Layers **1–3**: network support (physically moving data)
- Layer **5**: user support (interoperability between software)
- Layer **4** links the two groups and ensures the lower layers deliver data in a form the upper layers can use

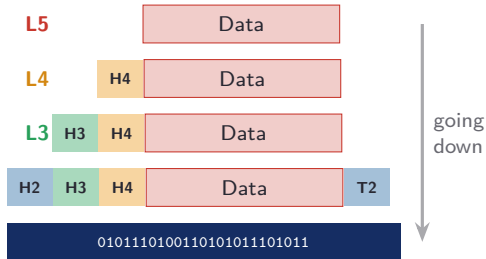
Design goal

A **comprehensive and flexible** architecture.

A Message Travels from Device A to Device B



Intermediate nodes (routers) usually involve **only the first three layers**. Between machines, layer x on one machine communicates with layer x on the other using a **protocol**.



- **Peer-to-peer processes:** the processes at the same layer on each machine
- Each sending layer **adds its own header** (and sometimes a trailer) and passes the package down
- At layer 1 everything becomes a **bit stream**
- The receiver **unwraps** layer by layer
- **Interfaces** between adjacent layers define what services each layer offers



1 Physical Layer

Transmits a **bit stream** over a physical medium.

- **Interface & media:** mechanical and electrical specifications
- **Representation of bits:** encode bits as electrical or optical signals
- **Data rate:** bits sent per second
- **Synchronization:** sender and receiver clocks agree



2 Data Link Layer

Turns the raw physical layer into a **reliable link**.

- **Framing:** divides the bit stream into **frames**
- **Physical addressing:** header holds sender and receiver addresses
- **Flow control:** does not overwhelm a slow receiver
- **Error control:** detects and retransmits damaged or lost frames (trailer)
- **Access control:** decides which device uses the link

Header	Data from layer 3	Trailer
--------	-------------------	---------



3 Network Layer

Source-to-destination delivery of a **packet**, possibly across many networks.

- **Logical addressing:** addresses that work beyond the local network (e.g. IP address)
- **Routing:** **routers** forward packets to their final destination

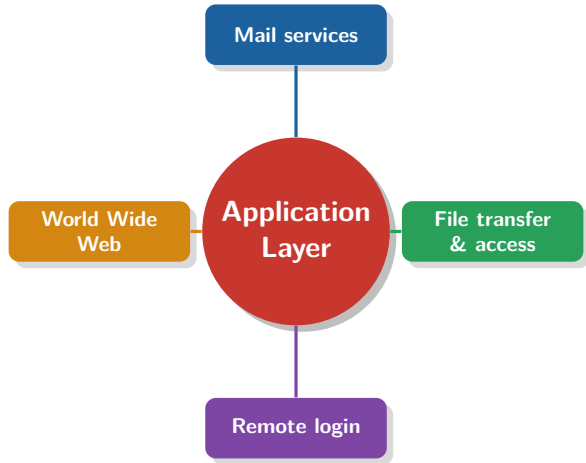


4 Transport Layer

Process-to-process delivery of the **entire message**, in order and intact.

- **Port addressing:** reach the right *program* on a computer
- **Segmentation & reassembly:** segments carry sequence numbers
- **Connection control:** connectionless or connection-oriented
- **Flow & error control:** end to end, not per link

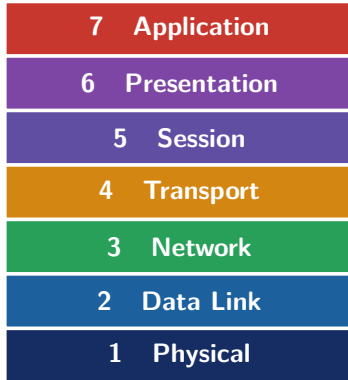




Lets the **user** (a person or software) **access the network**. It provides user interfaces and support for services:

- **Mail services**: e-mail forwarding and storage
- **File transfer & access**: retrieve and manage files on a remote host
- **Remote login**: log in and use a remote computer
- **WWW**: the most common application today

The OSI Model: Seven Layers



“All People Seem To Need Data Processing”

- **Open Systems Interconnection** model by the **ISO**
- A **theoretical** model showing how a protocol stack *should* be built
- Never seriously implemented as a real protocol stack
- Adds **two extra layers** to the Internet model:

Session layer (5)

The network **dialog controller**: establishes, maintains and synchronizes interaction.

Presentation layer (6)

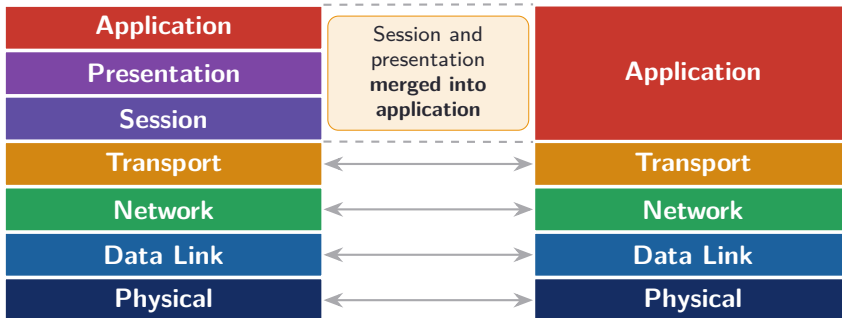
Syntax and semantics: **translation, encryption/decryption, compression.**

OSI vs. TCP/IP: Side by Side



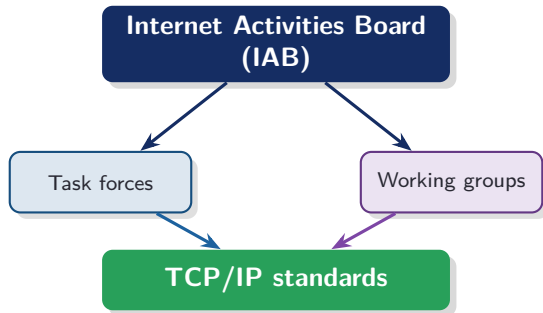
OSI (7 layers)

Internet / TCP/IP (5 layers)



Why the five-layer model is used more

Session and presentation duties are now handled by other layers: encryption happens at **several** layers, compression at the application layer.



- One of the most successful bodies for setting **data transfer standards** in the Internet arena
- Has **overall authority** for these activities
- Several task forces and groups **define new protocols** and **improve existing ones**
- Best-known result: the widely used **TCP/IP** suite



5 Application	 PGP, S/MIME, Kerberos, SET	Modules 11–12
4 Transport	 SSL/TLS (secure web)	Module 12
3 Network	 IPSec (AH, ESP)	Module 12
2 Data Link	 Link encryption	–
1 Physical	 Physical protection of cables	–

Big picture

The **cryptography** of Lesson 1 is the tool; the **layers** of Lesson 2 are where we apply it. The rest of the course builds on both.



Lesson 1: Introduction to Network Security

- CIA goals and the three aspects
- Interruption, interception, modification, fabrication
- Cryptology = cryptography + cryptanalysis
- 3 dimensions of cryptosystems
- Brute force and 6 cryptanalytic attacks

Lesson 2: Network Models

- LAN, MAN, WAN and internetworks
- Layered tasks (postal analogy)
- ISO and the OSI 7-layer model
- TCP/IP 5-layer Internet model
- Duties of each layer
- IAB and Internet standards

Review questions

1. Discuss cryptography in brief.
2. Define cryptanalysis.
3. Explain different categories of networks.
4. Differentiate between the OSI and TCP/IP models.



- ① William Stallings, *Cryptography and Network Security*, PHI Publishers.
- ② Charlie Kaufman, Radia Perlman, *Network Security*, Pearson Education.
- ③ Bruce Schneier, *Applied Cryptography*, John Wiley & Sons.
- ④ Roberta Bragg, *Network Security: The Complete Reference*, TMH.

Thank You! Questions?